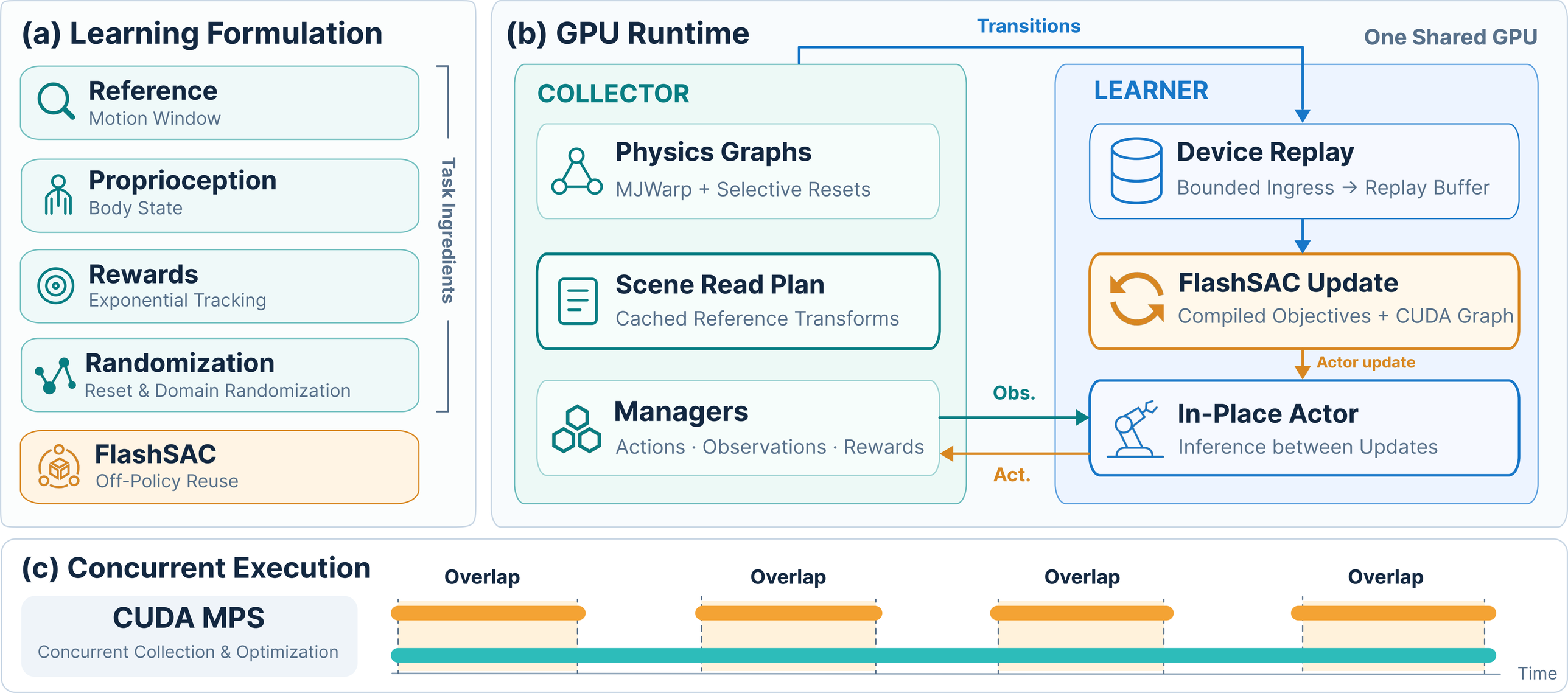

(a) Learning Formulation
Reference
Motion Window
Proprioception
Body State
Rewards
Exponential Tracking
Randomization
Reset & Domain Randomization
Task Ingredients
FlashSAC
Off-Policy Reuse
(b) GPU Runtime
Transitions
One Shared GPU
COLLECTOR
Physics Graphs
MJWarp + Selective Resets
Scene Read Plan
Cached Reference Transforms
Managers
Actions · Observations · Rewards
Obs.
Act.
LEARNER
Device Replay
Bounded Ingress → Replay Buffer
FlashSAC Update
Compiled Objectives + CUDA Graph
Actor update
In-Place Actor
Inference between Updates
(c) Concurrent Execution
CUDA MPS
Concurrent Collection & Optimization
Overlap
Overlap
Overlap
Overlap
Time